GUIDE TO CODES & DATA DICTIONARY

3RD EDITION — BOUHEDMA · Biodiversity Monitoring & Human-Wildlife Conflicts · September 14–19, 2026

WHY A CONTROLLED VOCABULARY?
Thirty participants, thirteen practical sessions, six days: without a common vocabulary, the final dataset will contain "jackal", "Golden Jackal", "C. aureus", and "jackal?" for the same species, rendering it unusable. Never invent a code in the field: if a code is missing, write the full species name in plain text and report it to the data manager in the evening.

01 · GENERAL DATA ENTRY RULES

Rule	Application	Mandatory Format
One row = one observation	Never enter two species on the same row, never write "3 gazelles + 1 oryx" in a single cell.	—
Never leave a cell blank	Write **NA** (not applicable) or **ND** (not determined). A blank cell is indistinguishable from an omission.	NA or ND
No units inside cells	Write "42" instead of "42 m". Measurement units belong in the column header.	Pure number
Decimal separator = period	Write `34.47512` and not `34,47512` — commas break CSV files.	Decimal point (`.`)
No color-coded meaning	A highlighted cell is not preserved in export. Contextual details must be entered in dedicated columns.	Explicit text
Zero ≠ empty	"0" means "searched for but not found". This is valuable data.	0
2H Graphite Pencil	Pen ink smudges from sweat and dew. Pencil allows clean corrections and lasts.	Wooden pencil
Record immediately on-site	A sheet filled from memory back at camp in the evening is an inaccurate sheet.	Immediate entry

DATES, TIMES, AND COORDINATES

Field	Mandatory Format	Example / Note
Date	`YYYY-MM-DD`	`2026-09-16` (ISO 8601). Never use DD/MM/YY.
Time	`HH:MM` 24-hour format	`19:45`. Time zone: UTC+1 (Tunisian local time), constant throughout the week.
Latitude / Longitude	Decimal degrees, WGS 84, 5 decimal places	e.g., `34.47512 / 9.51038`. 5 decimal places ≈ 1 m accuracy. Never mix with degrees-minutes-seconds.
GPS Precision	In meters, as displayed on the device at time of recording	e.g., 4.
Elevation	In meters, integer. Source: GPS (low vertical accuracy, note as such).	e.g., 153.
Waypoint	Identical name to the sampling unit identifier	e.g., BHD26_CT03.

PARK GPS SET TO DEGREES-MINUTES?
Before the first field session, configure all devices to `hddd.ddddd°` and datum **WGS 84**, then cross-check each device.

02 · CONTROLLED VOCABULARIES

Each vocabulary is independent: the same code can exist in two different vocabularies without ambiguity, as the column on the datasheet indicates which applies.

V1 · Habitat

Code	Habitat
SAV	Vachellia tortilis savanna (plain)
STA	Alfa grass steppe (Stipa tenacissima)
STC	Chamaephyte steppe (Wormwood, Gymnocarpos)
JEB	Rocky jebel slope, matorral
OUE	Wadi bed, riparian vegetation
GLA	Glacis, stony reg
SAB	Sandy deposit, nebkha
HAL	Saline area, halophytic vegetation
ENC	Reintroduction enclosure
PTE	Water point, trough
CUL	Cropland, garden, periphery
BAT	Built-up area, camp, infrastructure
PIS	Track, road

V2 · Substrate

Code	Substrate
SAB	Loose sand
LIM	Silt, fine compacted soil
GRV	Gravel, reg
ROC	Bedrock or boulders
GYP	Gypsum or caliche crust
ARG	Clay, cracked soil
LIT	Plant litter

V3 · Sign Type

Code	Sign
VIS	Sighted animal
AUD	Heard animal
EMP	Footprint / track
FEC	Dung / scats
LAT	Latrine
TER	Burrow / den
GRT	Scratch mark / digging
POI	Hair / fur
PLU	Feathers
OSS	Bones / prey remains
CAD	Carcass
PEL	Pellet (raptor)
MUE	Shed skin (reptile)
PHO	Camera trap photo

V7 · Human-Wildlife Interaction

Code	Recorded Interaction Type
PRE	Predation on domestic livestock
DEG	Crop or orchard damage
PAT	Livestock grazing inside protected area
COU	Wood cutting / harvesting of woody plants
PIE	Trap, snare, loop
TIR	Shooting sign (spent shell, hide)
CHI	Stray or herd dog
DEC	Waste / illegal dumping
VEH	Off-road vehicle driving
MOR	Anthropogenic mortality

Any sign coded PIE, TIR, or MOR must be reported the same evening to the park conservator. This is not training data.

03 · DATASET STRUCTURE — "EVENT / OCCURRENCE" MODEL

All datasheets from Edition 3 translate into two linked tables aligned with the Darwin Core standard used by GBIF. This corresponds to the practical exercise in Session D4-S2 · Data Structuring.

04 · CONTROLLED VOCABULARIES (CONTINUED)

V4 · Age and Sex

Code	Meaning
M / F / I	Male / Female / Undetermined sex
AD	Adult
SA	Subadult
JU	Juvenile
JA	Young of the year
IND	Undetermined age

V5 · Identification Reliability

Code	Meaning
1	**Certain** — diagnostic features seen, no confusable species
2	**Probable** — consistent, but one key criterion missing
3	**Possible** — several species remain plausible

Defaulting to genus level (*Gerbillus sp.*) is always better than guessing a species. An honest "3" is worth more than a false "1": the data remains usable.

V6 · Weather Conditions

Field	Coding
Cloud cover	In oktas, 0 to 8
Wind	Beaufort scale 0 to 12
Rain	0 none · 1 drizzle · 2 showers
Temperature	°C, integer
Moon phase	NM · FQ · FM · LQ (new moon, first quarter, full moon, last quarter)

Weather is recorded at the start and end of each sampling unit, not once for the whole day: bird, bat, and insect detectability is directly tied to it.

05 · NAMING CONVENTIONS FOR IDENTIFIERS AND FILES

Object	Template and Example
Camera trap station	`BHD26_CT01 ... CT04`
Acoustic station / point count	`BHD26_AC01 ... (acoustic)`
Transect	`BHD26_TR01 ... TR04`
Quadrat on transect	`BHD26_TR02_Q3` (3rd quadrat of transect 2)
Light trap station	`BHD26_LT01, LT02`
Botanical record	`BHD26_RB01 ...`
Herbarium specimen	`BHD26_H001 ...` continuous numbering
Sample (feces, insect, tick)	`BHD26_E001 ...` continuous numbering
Team	`A, B, C, D`
Observer	3-letter unique initials — e.g., ZBO. Allocation table kept at camp.

File Type	Template and Example
Session data entry	`BHD26_13S1_ENTRY_2026-09-16.csv`
Station photos	Folder `BHD26_CT01_2026-09-19/`
Audio recordings	Folder `BHD26_AC01_2026-09-15/`
Paper sheet scan	`BHD26_13S1_FT01_CT02_scan.jpg`

THREE FORBIDDEN PRACTICES IN FILE NAMING:
No spaces (use _) · no accents or special characters · no dates in DD-MM-YYYY. Original media files (photos, audio) are **never renamed, cropped, or edited**: work is always conducted on a duplicate copy.

THE GOLDEN RULE:
Effort is recorded even when nothing is observed. A camera-night with zero detections, a transect with no birds, a quadrat with no plants: these are legitimate rows in the Event table with zero associated occurrences. Without them, it is impossible to calculate detection probability, frequency, or density — rendering the week's field effort analytically invalid.

06 · SPECIES CODES — MAMMALS

4-letter alpha code: first two letters of genus + first two letters of specific epithet (*Oryx dammah* → ORDA). Any species missing from this list is written in full on the datasheet; the code will be assigned at camp. IUCN Red List status is indicative and must be verified at date of use.

Code	Scientific Name	English Name	Note
ORDA	*Oryx dammah*	Scimitar-horned Oryx	Reintroduced 1985 · enclosure
ADNA	*Addax nasomaculatus*	Addax	Reintroduced · enclosure
NADA	*Nanger dama mhorr*	Mhorr Dama Gazelle	Reintroduced 1990s
GADO	*Gazella dorcas*	Dorcas Gazelle	Free-ranging population
AMLE	*Ammotragus lervia*	Barbary Sheep	Jebel slopes
HYHY	*Hyaena hyaena*	Striped Hyena	Sensitive data
CAAU	*Canis aureus*	Golden Jackal	Livestock conflict
CACA	*Caracal caracal*	Caracal	Rare, elusive
FELY	*Felis lybica*	African Wildcat	Possible hybridization
VUVU	*Vulpes vulpes*	Red Fox	Sandy sectors
VURU	*Vulpes rueppellii*	Rüppell's Fox	Ubiquitous
HEIC	*Herpestes ichneumon*	Egyptian Mongoose	Diurnal
GEGE	*Genetta genetta*	Common Genet	To be confirmed locally
HYCR	*Hystrix cristata*	Crested Porcupine	Quills on ground
LECA	*Lepus capensis*	Cape Hare	Common
CTGU	*Ctenodactylus gundi*	Common Gundi	Jebel rocks
JAJA	*Jaculus jaculus*	Lesser Egyptian Jerboa	Nocturnal
PSOB	*Psammomys obesus*	Fat Sand Rat	Diurnal, on Salsolaceae
MESH	*Meriones shawii*	Shaw's Jird	Burrowing colonies
GERSP	*Gerbillus sp.*	Gerbil species	Genus level — hand ID required
ATAL	*Atelerix algirus*	Algerian Hedgehog	Nocturnal
PAAE	*Paraechinus aethiopicus*	Desert Hedgehog	Nocturnal

Bats — Working Codes

Code	Scientific Name	English Name
RHCY	*Rhinopoma cystops*	Egyptian Mouse-tailed Bat
ASTR	*Asellia tridens*	Trident Bat
TANU	*Taphozous nudiventris*	Naked-rumped Tomb Bat
OTHE	*Otonycteris hemprichii*	Desert Long-eared Bat
PIKU	*Pipistrellus kuhlii*	Kuhl's Pipistrelle
PIRU	*Pipistrellus rueppellii*	Rüppell's Pipistrelle
HYSA	*Hypsugo savii*	Savi's Pipistrelle
EPIS	*Eptesicus isabellinus*	Isabelline Serotine
PLGA	*Plecotus gaisleri*	Maghreb Long-eared Bat
TATE	*Tadarida teniotis*	European Free-tailed Bat
RHHI	*Rhinolophus hipposideros*	Lesser Horseshoe Bat
NYTH	*Nycteris thebaica*	Egyptian Slit-faced Bat
CHISP	*Unidentified Bat*	Default code

BATS: DEFAULT CODE IS THE RULE. No bat species identification is validated in the field based solely on real-time acoustic listening. On the datasheet, record CHISP alongside acoustic parameters (peak frequency, call structure, duration, pulse rate). Species identification is performed post-field by trainers analyzing sonograms against regional reference libraries. This list is for reviewing final outputs, **not for field acoustic identification**.

SENSITIVE SPECIES DATA: Striped Hyena, Caracal, Barbary Sheep, and all poaching indicators: coordinates are not publicly disclosed and must not

07 · BIRDS — EXPECTED SPECIES IN SEPTEMBER

Non-exhaustive working checklist focusing on resident species and mid-September post-breeding migrants. Any species off-list must be written in full text.

Code	Scientific Name	English Name
STCA	*Struthio camelus*	North African Ostrich (reintroduced)
AQFA	*Aquila fasciata*	Bonelli's Eagle
NEPE	*Neophron percnopterus*	Egyptian Vulture
FATI	*Falco tinnunculus*	Common Kestrel
BUAS	*Bubo ascalaphus*	Pharaoh Eagle-Owl
ATNO	*Athene noctua*	Little Owl
ALBA	*Alectoris barbara*	Barbary Partridge
PTAL	*Pterocles alchata*	Pin-tailed Sandgrouse
CUCU	*Cursorius cursor*	Cream-colored Courser
CORU	*Corvus ruficollis*	Brown-necked Raven

Code	Scientific Name	English Name
GACR	*Galerida cristata*	Crested Lark
GATH	*Galerida theklae*	Thekla's Lark
AMDE	*Ammomanes deserti*	Desert Lark
OELE	*Oenanthe leucura*	Black Wheatear
OEDE	*Oenanthe deserti*	Desert Wheatear
LAME	*Lanius meridionalis*	Iberian Grey Shrike
BUGI	*Bucanetes githagineus*	Trumpeter Finch
EMSA	*Emberiza sahari*	House Bunting
COLI	*Columba livia*	Rock Dove
AVISP	*Unidentified Bird*	Default code

08 · REPTILES AND AMPHIBIANS

Code	Scientific Name	English Name	Note
VAGR	*Varanus griseus*	Desert Monitor	CITES App. I
URAC	*Uromastyx acanthinura*	Spiny-tailed Lizard	CITES App. II
TEGR	*Testudo graeca*	Spur-thighed Tortoise	CITES App. II
CHCH	*Chamaeleo chamaeleon*	Common Chameleon	CITES App. II
CECE	*Cerastes cerastes*	Sahara Horned Viper	Venomous
CEVI	*Cerastes vipera*	Sahara Sand Viper	Venomous
DADE	*Daboia mauritanica*	Moorish Viper	Venomous
MAIN	*Malpolon insignitus*	Eastern Montpellier Snake	Opisthoglyphous
PSSC	*Psammophis schokari*	Schokari Sand Racer	Opisthoglyphous
AGIM	*Agame impatialis*	Bibron's Agama	Rocky outcrops
ACASP	*Acanthodactylus sp.*	Fringe-toed Lizard	Genus — scale counts required
MESSP	*Mesalina sp.*	Desert Lacerta	Genus
CHOC	*Chalcides ocellatus*	Ocellated Skink	—
PESA	*Pelophylax saharicus*	Sahara Frog	Water points

HANDLING SAFETY: No snakes are handled during the Academy, whether venomous or non-venomous. Observation, photography, and distance measurements only. CITES-listed species must not be captured, collected, or translocated: possession and transport are subject to authorization from the DGF (CITES Management Authority).

09 · FLORA — KEY STRUCTURAL SPECIES OF BOUHEDMA

Taxonomy checked against the *Catalogue synonymique commenté de la flore de Tunisie* (Le Floc'h, Boulos & Véla, 2010) and POWO. Common synonyms are provided.

Code	Scientific Name	Type / Zone
VATO	*Vachellia tortilis subsp. raddiana var. silvestris*	Tree — savanna (protected)
RHTR	*Rhus tripartita*	Shrub — glacis, jebel
ZILO	*Ziziphus lotus*	Shrub — steppe
PEAN	*Periploca angustifolia*	Shrub — plain
RERA	*Retama raetam*	Shrub — sand dunes, wadis
PIAT	*Pistacia atlantica*	Tree — wadis, jebel
OLEU	*Olea europaea*	Tree — jebel
JUPH	*Juniperus phoenicea*	Tree — jebel slopes
STTE	*Stipa tenacissima*	Grass — alfa steppe
LYSP	*Lygeum spartum*	Grass — saline soils
STPU	*Stipagrostis pungens*	Grass — sand dunes
CECI	*Cenchrus ciliaris*	Grass — acacia understory

Code	Scientific Name (Hammada) / (Pituranthos)	Type / Zone
ARHE	*Artemisia herba-alba*	Chamaephyte — steppe
ARCA	*Artemisia campestris*	Chamaephyte — sands
GYDE	*Gymnocarpos decander*	Chamaephyte — glacis
ANHE	*Anthyllis henoniana*	Chamaephyte — steppe
HEKA	*Helianthemum kahiricum*	Chamaephyte — glacis
ASAR	*Astragalus armatus*	Spiny chamaephyte — overgrazed
THHI	*Thymelaea hirsuta*	Chamaephyte — degraded areas
ATHA	*Atriplex halimus*	Halophyte
SAVE	*Salsola vermiculata*	Halophyte
HASO	*Haloxylon scoparium*	Halophyte — reg
LIGU	*Limoniastrum guyonianum*	Halophyte — gypseous soils
NIRE	*Nitraria retusa*	Halophyte — nebkhas
DETO	*Deverra tortuosa*	Chamaephyte — wadis
CASP	*Capparis spinosa*	Shrub — rocky cliffs

10 · DATA WORKFLOW & LIFECYCLE AFTER THE FIELD WEEK

1. Paper datasheet filled on-site in the field, counter-signed by quality controller.
2. Datasheet photographed the same evening (immediate backup prior to digital entry).
3. Digitized into the CSV session template by the designated team data entry pair.
4. Cross-verification: entry re-read and validated by a secondary team.
5. Consolidation into Event / Occurrence schema (Session D4-S2 workshop).
6. Official handover to the Bouhedma NP Conservator and DGF (data owners).

DATA OWNERSHIP AND DISSEMINATION: Data collected within a protected area belongs exclusively to the managing authority of that area. Participants shall not publish, distribute, or upload field data to public platforms without written consent from the Park Conservator and the DGF. Personal photographs remain the property of their author, but sensitive species locations must never be shared or geotagged.